Job Market Analytics Dashboard

Step 6 — Plotly Interactive Charts

Step 6 Overview

Status: ✓ Complete

Installed Plotly, added three interactive charts to the dashboard — Top Skills in Demand (horizontal bar), Top Hiring Locations (vertical bar), and Salary Distribution (histogram). All charts use the dark theme color palette matching the portfolio aesthetic. Charts are fully interactive with zoom, pan, and hover.

6.1 Install Plotly

```
pip install plotly
pip freeze > requirements.txt
```

6.2 New Imports Added to app.py

```
import plotly.express as px
import plotly
import json
```

Import	What it does
plotly.express as px	High-level chart API — px.bar(), px.histogram() etc.
plotly	Core library — needed for PlotlyJSONEncoder
json	Converts Plotly figure to JSON string for embedding in HTML

6.3 Chart 1 — Top Skills in Demand (Horizontal Bar)

```
# Build data dict from skill_counts list of tuples
skills_df_data = {
"Skill": [s[0] for s in skill_counts],
"Jobs": [s[1] for s in skill_counts]
}
# Create horizontal bar chart
fig_skills = px.bar(skills_df_data, x="Jobs", y="Skill", orientation="h",
title="Top Skills in Demand",
color_discrete_sequence=["#c9a84c"]) # gold accent color
# Apply dark theme
fig_skills.update_layout(
paper_bgcolor="#0d0d0d", # outer background
plot_bgcolor="#1a1a1a", # chart area background
font_color="#e8e6df", # axis labels
title_font_color="#c9a84c",
yaxis=dict(autorange="reversed") # highest count at top
)
# Convert to JSON for embedding in HTML
chart_skills = json.dumps(fig_skills, cls=plotly.utils.PlotlyJSONEncoder)
```

6.4 Chart 2 — Top Hiring Locations (Vertical Bar)

```
loc_data = {
"Location": [l[0] for l in location_counts],
"Jobs": [l[1] for l in location_counts]
}
fig_loc = px.bar(loc_data, x="Location", y="Jobs",
title="Top Hiring Locations",
color_discrete_sequence=["#1D9E75"]) # teal color
fig_loc.update_layout(paper_bgcolor="#0d0d0d", plot_bgcolor="#1a1a1a",
```

```
font_color="#e8e6df", title_font_color="#1D9E75")
chart_location = json.dumps(fig_loc, cls=plotly.utils.PlotlyJSONEncoder)
```

6.5 Chart 3 — Salary Distribution (Histogram)

```
# Extract only jobs that have salary data
salary_data = [j["salary_min"] for j in jobs if j.get("salary_min")]
fig_salary = px.histogram(x=salary_data, nbins=15,
title="Salary Distribution",
labels={"x": "Salary (£)"},
color_discrete_sequence=["#7F77DD"]) # purple color
fig_salary.update_layout(paper_bgcolor="#0d0d0d", plot_bgcolor="#1a1a1a",
font_color="#e8e6df", title_font_color="#7F77DD")
chart_salary = json.dumps(fig_salary, cls=plotly.utils.PlotlyJSONEncoder)
```

6.6 How Charts Embed in HTML (dashboard.html)

Step 1 — Load Plotly JS from CDN in the head:

```
<script src="https://cdn.plot.ly/plotly-latest.min.js">
```

Step 2 — Create div containers for each chart:

```
<div id="chart-skills">
<div id="chart-location">
<div id="chart-salary">
```

Step 3 — Initialize charts with JavaScript at bottom of page:

```
<script>
Plotly.newPlot("chart-skills",
{{ chart_skills | safe }}.data,
{{ chart_skills | safe }}.layout);
Plotly.newPlot("chart-location",
{{ chart_location | safe }}.data,
{{ chart_location | safe }}.layout);
Plotly.newPlot("chart-salary",
{{ chart_salary | safe }}.data,
{{ chart_salary | safe }}.layout);
</script>
```

The | safe filter tells Jinja2 to render the JSON as raw HTML without escaping special characters like < > & — necessary for JavaScript to read it correctly.

6.7 Chart Summary

Chart	Type	Color	Data source	Interactive features
Top Skills in Demand	Horizontal bar	Gold #c9a84c	skill_counts top 10	Hover count, zoom, pan, download
Top Hiring Locations	Vertical bar	Teal #1D9E75	location_counts top 5	Hover count, zoom, pan, download
Salary Distribution	Histogram	Purple #7F77DD	salary_min all jobs	Hover range, zoom, bin adjustment

6.8 Result in Browser

Dashboard now shows:

Section	Content
Stat cards	60 Jobs Tracked | £64,937 Average Salary | Machine Learning (Top Skill)
Skills chart	Horizontal bars — machine learning 25, python 22, data science 10...
Locations chart	Vertical bars — London UK 41, The City Central London 16...
Salary histogram	Distribution of salary_min across all 60 jobs

Recent jobs	5 job cards with title, company, salary, skill tags, View Job link

Result: ✓ Three interactive Plotly charts rendered with dark theme. Dashboard looks professional.

Build Progress

Step	Description	Status
1	Project setup — venv, packages, file structure, local Flask server	✓ Done
2	Adzuna API — register, get keys, write fetch_jobs.py, pull live data	✓ Done
3	Data cleaning — Pandas, normalize fields, extract skill tags	✓ Done
4	Supabase — create jobs table, store first batch of 60 jobs	✓ Done
5	Dashboard route — query Supabase, compute analytics, render template	✓ Done
6	Plotly charts — skills, location, salary interactive charts	✓ Done
7	Jobs listing page — /jobs route with filters	Next →
8	UI polish — dark CSS theme	
9	GitHub + Railway deployment	
10	Add to portfolio via /admin	
11	(Stretch) ML salary prediction model	
